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Medi Track: A Hospital Record Management System

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Medi Track

(Hospital Records Management System)

Introduction

Overview of the System

Medi Track is a desktop application designed to help hospitals digitally monitor their internal activities, particularly patient records. It streamlines record-keeping, ensuring efficient and organized management of hospital data. It allows hospital medical personnel to have access to information on each patient through this digital monitoring system, making it easier for them to continue providing efficient and accurate healthcare services. The application includes login systems for different medical personnel each with different level of access control based on their roles. The system also includes features for appointment scheduling and medical history records for efficient monitoring service. With its user-friendly interface and secure database integration, Medi Track ensures that hospital operations run smoothly while maintaining the highest standards of patient care and data confidentiality.

Reasons for the development of the System

Health care is the lifeblood of society thus, investing in its development is crucial for the survivability of a country and its people. However, there are some health care establishments in the Philippines that still uses the traditional method of monitoring the records of the patients which causes human errors and delays in the information access. In fact, there are hospitals that still utilizes paper records and often causes long wait times, time-consuming record retrieval, data loss and misplacement, error data entries and even data insecurity. It is very important for our health care systems to go along with the innovation so that it can sustain and continue to provide quality service to the people especially with the growing population. According to the Institute of Electrical and Electronic Engineers (IEEE), healthcare monitoring systems in hospitals and other health facilities have grown significantly, and portable healthcare monitoring systems using emerging technologies such as Internet of Things (IoT) technologies have aided the advancement of healthcare monitoring systems. With this, the developers want to take this issue as an opportunity to challenge their selves and their capabilities by making a crucial step in attempt to make an initiatory action through developing this system.

Specifications or Features Finished

Medi Track has implemented a range of features designed to significantly enhance the efficiency of the system. These improvements focus on streamlining operations and optimizing workflows, ensuring a more seamless and effective user experience. Below are the list of features of the system.

1.Login System

A secure and user-friendly login system designed for medical personnel, including nurses, doctors, receptionists, and admins, to access their accounts. This feature ensures that only authorized users can log in and access the platform. Users can either log into their existing accounts or create a new one by providing their credentials. The system helps maintain data security and ensures that only staff with the proper permissions can access sensitive hospital information.

2.Appointment System

This feature allows medical personnel, particularly receptionists, to schedule appointments for patients and assign them to the appropriate doctor. It includes the appointment booking process, ensuring that patients are matched with the correct healthcare professional based on their needs.

3. Add Patients

This feature allows medical personnel to add new patients to the system, ensuring that all relevant personal and medical information is accurately recorded. It simplifies the process of patient registration, allowing healthcare providers to quickly update and manage patient data as needed.

4. QR Code generator

To be defined

5. Patient Records

This feature enables medical personnel to easily view and access patient records, ensuring that healthcare providers have quick and accurate information at their fingertips. It simplifies the process of retrieving patient's data such as their personal details and medical histories. It also enables the to see the number of patients in the hospital as well as their status. They can also updates patient data with this feature.

6. Billing System

This feature allows medical personnel to calculate and summarize all charges that a patient is required to pay. It simplifies the billing process, ensuring accurate bills by automatically adding up all medical services, treatments, and other related costs for each patient.

7. QR Code Reader

To be defined

8. Update System

The Update System allows medical personnel to modify existing records, ensuring accurate and up-to-date information across the platform. It enables staff to update patient details, appointments, billing records, and user accounts as needed. This feature helps maintain data integrity by allowing corrections, rescheduling, and adjustments in real time, ensuring efficient hospital management and patient care.

9. Delete System

The Delete System allows medical personnel to remove outdated or incorrect records from the platform, ensuring that only relevant and necessary information is retained. This feature enables staff to delete patient records, appointments, billing entries, and user accounts when required, helping maintain an organized and clutter-free database while upholding data security and compliance with hospital policies.

//Gi arrange ni nako sa according to crud

// to be defined pa ang QR since not sure pa hahaha

Tech Stack

The Medi Track desktop application is indeed an impactful system thus, developing this application is a challenge among the developers. With this, different tools were used to develop the said system. From designing its user interface, designing of database and writing the whole program, the developers carefully selected the right tools to ensure the efficiency and functionality of the app. Below is the list of technologies used during the development of the application.

Figma – Used for designing the user interface.

Visual Studio 2022 – An integrated development environment (IDE) from Microsoft, used for writing, debugging, and testing programs.

GitHub – A platform for version control, collaboration, and efficient management of code changes.

C# – A modern, object-oriented programming language developed by Microsoft, widely used for building Windows desktop applications, web apps, and games.

.NET WinForms – A powerful software development framework by Microsoft that provides libraries, tools, and runtime environments for building and running Windows applications.

ChatGPT – An AI language model that provides guidance, troubleshooting support, and technical solutions, including best practices and code suggestions.

MySQL – An open-source relational database management system (RDBMS) used to store and manage structured data in tables.

Krypton Toolkit - a UI framework for Windows Forms applications. It provides enhanced controls and themes to create modern and visually appealing desktop applications.

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References: <https://ieeexplore.ieee.org/abstract/document/10439065>